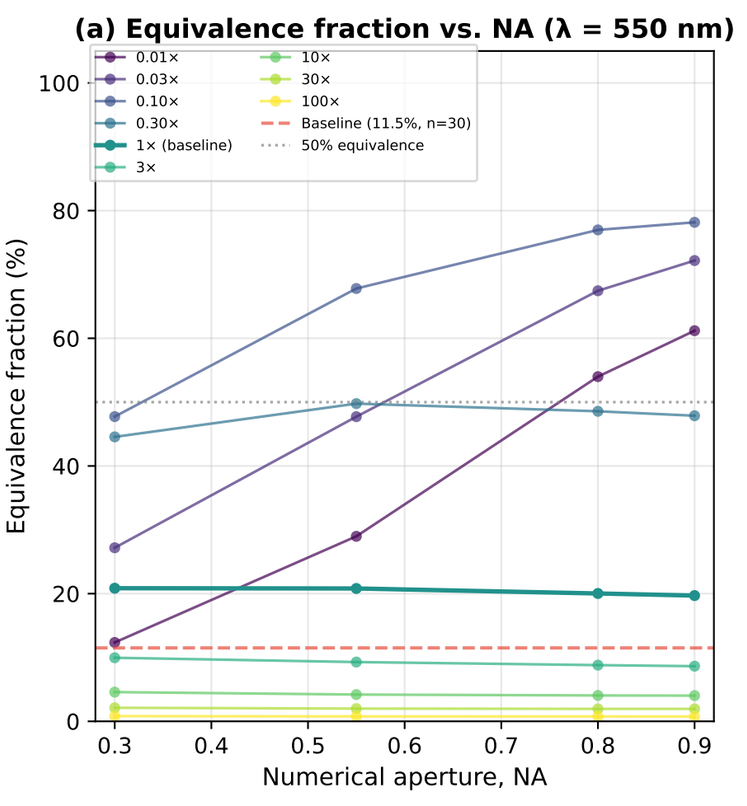
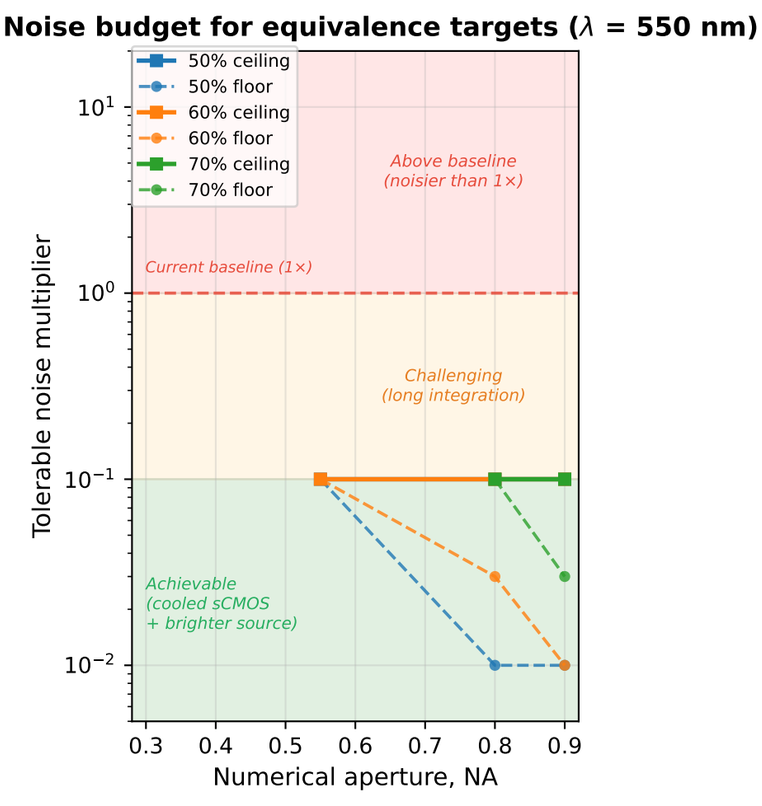




(c) Four improvements to close the gap

1. Reduce optical noise floor (**10 ×** sweep optimum)
Cooled sCMOS, brighter LED/laser, multi-frame averaging ($\sqrt{N}$ gain)
→ Single most impactful change

2. Increase NA to **0.7–0.9**
High-NA Mirau objectives (NA 0.8)
→ f_c from 2000 to 2909 mm^{-1} (2NA/λ)

3. Shorter **λ = 400–450** nm
Blue LED / diode laser
→ f_c reaches 4000 mm^{-1} (NA 0.8, λ=400 nm)

4. 2D phase unwrapping
Quality-guided or temporal PSI
→ 1% → 0.1% error rate

11.5% → 82.4% equivalence (sweep) -- **17%** residual: bias-limited